

Infinitesimal Calculus 2, Moed B

2025, Reichman university.

Do not write at the back of the papers!!

Lecturer: Dr. Yossi Shamaï. Duration: 3 hours.

Instructions: The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits. There is no choice between different items of the same question. **Unexplained answers will not receive any credit. You must state all of the conditions for every theorem and/or claim. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

1. (25 points)

1.1. (17 points) Let $0 \leq k \in \mathbb{N}$ and let $(a_n)_{n=k}^{\infty}, (b_n)_{n=k}^{\infty}$ be two sequences of positive numbers.

1.1.1. (8 points) Suppose that

$$\exists N \in \mathbb{N} : \forall n \geq N, \quad \frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}$$

Prove that $\sum_{n=N}^{\infty} b_n < \infty \Rightarrow \sum_{n=N}^{\infty} a_n < \infty$.

Solution: See lectures.

1.1.2. (9 points) Suppose that there exists an $L \in \mathbb{R}$ such that $0 \leq L < 1$ and such that

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$$

Prove that $\sum_{n=k}^{\infty} a_n < \infty$.

Solution:

See lectures.

1.2. (8 points) Prove or disprove the following statement: The following series is convergent:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{\sqrt{n}} + \frac{(-1)^n}{n} \right)$$

Solution:

False. We show that the series is divergent: note that

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{\sqrt{n}} + \frac{(-1)^n}{n} \right) = \sum_{n=1}^{\infty} \left(\frac{(-1)^{n+1}}{\sqrt{n}} - \frac{1}{n} \right) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}} - \sum_{n=1}^{\infty} \frac{1}{n}$$

The first series converges by the Leibniz test, whereas the second series diverges, as it is the Harmonic series. Therefore the series $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{\sqrt{n}} + \frac{(-1)^n}{n} \right)$ is divergent (convergent - divergent).

2. (25 points)

2.1. (15 points) Let $x_0 \in \mathbb{R}$ and let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers. Let $x_1, x_2 \in \mathbb{R}$ such that $|x_2 - x_0| < |x_1 - x_0|$. Suppose that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges at $x = x_1$. Prove that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely at $x = x_2$.

Solution:

See lectures.

2.2. (10 points) Find the domain of convergence of the power series:

$$\sum_{n=1}^{\infty} x^{2^n} = x^2 + x^4 + x^8 + x^{16} + x^{32} + \dots$$

Solution:

We note that for every $0 \leq x < 1$ we have

$$\sum_{n=1}^{\infty} x^{2^n} = x^2 + x^4 + x^8 + x^{16} + x^{32} + \dots \leq x + x^2 + x^3 + \dots = \sum_{n=1}^{\infty} x^n$$

Thus, by the comparison test, the series

$$\sum_{n=1}^{\infty} x^{2^n} = x^2 + x^4 + x^8 + x^{16} + x^{32} + \dots$$

converges for every $0 \leq x < 1$.

For $x = 1$:

We get the series

$$1 + 1 + 1 + \dots = \sum_{n=1}^{\infty} 1$$

As the necessary criterion does not hold, the series diverges.

For $x = -1$:

We get the series

$$1 + 1 + 1 + \dots = \sum_{n=1}^{\infty} 1$$

As the necessary criterion does not hold, the series diverges.

Thus, the domain of convergence is $-1 < x < 1$.

3. (25 points)

3.1. (13 points) Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences of real numbers. Suppose that $\sum_{n=1}^{\infty} a_n$ converges conditionally, and that $\sum_{n=1}^{\infty} b_n$ converges absolutely. Prove that $\sum_{n=1}^{\infty} (a_n - b_n)$ converges conditionally.

Solution:

First, as $\sum_{n=1}^{\infty} b_n$ converges absolutely then it converges, which implies that by the Algebra of series, $\sum_{n=1}^{\infty} (a_n - b_n)$ is convergent, as it is a difference of two convergent series. To prove that it converges conditionally, we must prove that it does not converge absolutely. ATC that $\sum_{n=1}^{\infty} |a_n - b_n| < \infty$. Then by the comparison principle for nonnegative series,

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |a_n - b_n + b_n| \stackrel{\Delta}{\leq} \sum_{n=1}^{\infty} (|a_n - b_n| + |b_n|) = \sum_{n=1}^{\infty} |a_n - b_n| + \sum_{n=1}^{\infty} |b_n| < \infty$$

in contradiction to the assumption that $\sum_{n=1}^{\infty} a_n$ converges conditionally.

3.2. (12 points) Find all the values of $a \geq 0$ for which the following series converges:

$$\sum_{n=2}^{\infty} \frac{\sqrt[n]{n^2} \cdot a^n}{n^{1.5}}$$

Solution:

We use the root test,

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{\sqrt[n]{n^2} \cdot a^n}{n^{1.5}}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n^2} \cdot a}{(\sqrt[n]{n})^{1.5}} = a$$

It follows that the series converges for $0 \leq a < 1$ and diverges for $a > 1$. For $a = 1$ the series is $\sum_{n=2}^{\infty} \frac{\sqrt[n]{n^2}}{n^{1.5}}$, and it converges with the 1.5-Harmonic series as

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\left[\frac{\sqrt[n]{n^2}}{n^{1.5}} \right]}{\left[\frac{1}{n^{1.5}} \right]} = \lim_{n \rightarrow \infty} (\sqrt[n]{n})^2 = 1$$

Thus, the series is convergent if and only if $0 \leq a \leq 1$.

4. (25 points)

4.1. (13 points)

4.1.1. (5 points) Compute the definite integral $\int_0^{\frac{\pi}{4}} \tan(x) dx$. Hint: Explain why the integral is well-defined.

Solution:

We have

$$\int_0^{\frac{\pi}{4}} \tan(x) dx = \int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos(x)} dx \stackrel{t=\cos(x) \Rightarrow dt = -\sin(x)dx}{=} - \int_1^{\frac{\sqrt{2}}{2}} \frac{dt}{t} = -[\ln|t|]_1^{\frac{\sqrt{2}}{2}} = -\ln\left(\frac{\sqrt{2}}{2}\right) = \frac{1}{2} \ln(2)$$

4.1.2. (8 points) Conclude that

$$\frac{\ln(2)}{2e} \leq \int_0^{\frac{\pi}{4}} e^{-x} \tan(x) dx \leq \frac{\ln(2)}{2}$$

Hint: Explain why the integral is well-defined. Note: you may assume that $\pi < 4$.

Solution:

Note that the function $e^{-x} \tan(x)$ is continuous on $\left[0, \frac{\pi}{4}\right]$ by AOC, and therefore the definite integral $\int_0^{\frac{\pi}{4}} e^{-x} \tan(x) dx$ exists. Consider the function defined by $f(x) = e^{-x}$ for every $x \in \left[0, \frac{\pi}{4}\right]$. Note that f is decreasing on $\left[0, \frac{\pi}{4}\right]$ and therefore

$$\frac{1}{e^{\frac{\pi}{4}}} = f\left(\frac{\pi}{4}\right) \leq f(x) \leq f(0) = 1 \quad \forall x \in \left[0, \frac{\pi}{4}\right]$$

multiplying the above inequality by $\tan(x) \geq 0$ for every $x \in \left[0, \frac{\pi}{4}\right]$ we obtain

$$\frac{\tan(x)}{e^{\frac{\pi}{4}}} \leq e^{-x} \tan(x) \leq \tan(x) \quad \forall x \in \left[0, \frac{\pi}{4}\right]$$

Integrating all sides of the inequality, then by the monotonicity of the definite integral the inequalities are preserved.

Using item 4.1.1 we conclude that

$$\frac{\ln(2)}{2e} \stackrel{\pi < 4}{\leq} \frac{\ln(2)}{2e^{\frac{\pi}{4}}} = \frac{1}{e^{\frac{\pi}{4}}} \int_0^{\frac{\pi}{4}} \tan(x) dx \leq \int_0^{\frac{\pi}{4}} e^{-x} \tan(x) dx \leq \int_0^{\frac{\pi}{4}} \tan(x) dx = \frac{\ln(2)}{2}$$

4.2. (12 points) Determine whether the following series converges or diverges:

$$\sum_{n=2}^{\infty} \left[\frac{\sin(n)}{\ln(n)} \cdot \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{5^n \cdot n!} \right]$$

Solution:

We check absolute convergence:

$$\sum_{n=2}^{\infty} \left| \frac{\sin(n)}{\ln(n)} \cdot \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{5^n \cdot n!} \right| \leq \sum_{n=1}^{\infty} \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{\ln(n) \cdot 5^n \cdot n!}$$

We now use the ratio test:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{\left[\frac{1 \cdot 4 \cdot 7 \cdots (3n-2) \cdot (3n+1)}{5^{n+1} \cdot (n+1)! \cdot \ln(n+1)} \right]}{\left[\frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{5^n \cdot n! \cdot \ln(n)} \right]} = \lim_{n \rightarrow \infty} \frac{1 \cdot 4 \cdot 7 \cdots (3n-2) \cdot (3n+1) \cdot 5^n \cdot n! \cdot \ln(n)}{5^{n+1} \cdot (n+1)! \cdot 1 \cdot 4 \cdot 7 \cdots (3n-2) \cdot \ln(n+1)} \\ &= \lim_{n \rightarrow \infty} \frac{(3n+1) \cdot \ln(n)}{5(n+1) \cdot \ln(n+1)} = \lim_{n \rightarrow \infty} \frac{\left(3 + \frac{1}{n}\right) \cdot \ln(n)}{\left(5 + \frac{5}{n}\right) \cdot \ln(n+1)} \stackrel{(\star)}{=} \frac{3}{5} < 1 \end{aligned}$$

where

$$\lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(n+1)} \stackrel{\text{Heine}}{=} \lim_{x \rightarrow \infty} \frac{\ln(x)}{\ln(x+1)} \stackrel{\frac{2}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{\left[\frac{1}{x}\right]}{\left[\frac{1}{1+x}\right]} = \lim_{x \rightarrow \infty} \frac{1+x}{x} \stackrel{\frac{2}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{1}{1} = 1 \quad (\star)$$

Thus, the series converges absolutely.

5. (25 points)

5.1. (7 points) Let f be a function that is continuously differentiable on the interval $[0, 1]$. Suppose that $f(1) = 2$ and that $\int_0^1 f(x) dx = 1$. Compute the value of the definite integral $\int_0^1 [x \cdot f'(x)] dx$.

Hint: Explain why the integral is well-defined.

Solution:

First, as f is continuously differentiable and as any continuous function defined in a closed interval is integrable on this interval, then $\int_0^1 [x \cdot f'(x)] dx$ exists. Integrating by parts, we get the following:

$$\int_0^1 [x \cdot f'(x)] dx = [xf(x)]_0^1 - \int_0^1 f(x) dx = f(1) - \int_0^1 f(x) dx = 2 - 1 = 1.$$

5.2. (18 points) Let $(a_n)_{n=1}^{\infty}$ be a strictly increasing sequence of natural numbers.

5.2.1. (10 points) Let $p \in \mathbb{R}$. Suppose that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{a_n^p}$ converges conditionally. Prove that $0 < p \leq 1$.

Solution:

As the sequence $(a_n)_{n=1}^{\infty}$ is a strictly increasing sequence of natural numbers, then we know that $a_n \geq n$ for every $n \in \mathbb{N}$.

ATC first that $p \leq 0$. As $-p \geq 0$ then

$$\left| \frac{(-1)^{n+1}}{a_n^p} \right| = a_n^{-p} \geq 1$$

Thus, $\lim_{n \rightarrow \infty} \frac{(-1)^{n+1}}{a_n^p} \neq 0$ and therefore, by the necessary criterion, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{a_n^p}$ is divergent. Contradiction.

ATC now that $p > 1$. Note that

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{a_n^p} \right| = \sum_{n=1}^{\infty} \frac{1}{a_n^p} \stackrel{a_n \geq n}{\leq} \sum_{n=1}^{\infty} \frac{1}{n^p} \stackrel{p\text{-Harmonic}}{<} \infty$$

It follows that the series converges absolutely and therefore, does not converge conditionally. Contradiction.

5.2.2. (8 points) Give an example for a strictly increasing sequence of natural numbers $(a_n)_{n=1}^{\infty}$ for which the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{a_n^p}$ does not converge conditionally for every $p \in \mathbb{R}$.

Solution:

Consider the sequence defined by $a_n = 2^n$ for every $n \in \mathbb{N}$ and let $p \in \mathbb{R}$. ATC that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{a_n^p}$ converges conditionally. By item 5.2.1, we have $0 < p \leq 1$. We now show that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{a_n^p}$ converges absolutely.

Note that

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(-1)^{n+1}}{a_n^p} \right|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{2^{pn}}} = \frac{1}{2^p} \stackrel{p > 0}{<} 1$$

Thus, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{a_n^p}$ converges absolutely by the root test. Contradiction.